

Week 4 Tutorial Questions

Introduction

Recall the following definitions of 1-period returns from the Week 2 lecture.

Definition 1

Let P_t be the price of an asset at time t . The **net return** over the period $t - 1$ to t is

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

Definition 2

The simple **gross return** is defined as

$$1 + R_t = \frac{P_t}{P_{t-1}}.$$

Definition 3

Continuously compounded return, or **log return**, is denoted by r_t and defined as

$$r_t = \log(1 + R_t) = \log\left(\frac{P_t}{P_{t-1}}\right) = \log(P_t) - \log(P_{t-1}).$$

Implicitly, returns measure the appreciation of an asset over a holding period (1-period in this case). As such, we want returns to be positive and large. Returns are a meaningful measure of investor profit.

Question 1

Let r_t be log-returns of an asset. Suppose that $r_1, r_2, \dots, r_t$ are independent and identically distributed (i.i.d.) as normal random variables with mean μ and variance σ^2 . Using your knowledge of the normal distribution, answer the following questions.

1. What is the distribution of the 2-period return $r_t(2) = r_t + r_{t-1}$? Show your working.

Note: If this was an assignment, you have the option to NEATLY hand-write the answer and insert it here as an answer. You need to make sure that when you save your complete file to pdf, that your answer is legible and not too small

Hint: The sum of two normally distributed variables is normal

2. Let $\mu = 0.1$ and $\sigma^2 = 0.6$. Use R to find $\Pr[r_t(2) < 2]$.
3. What is the covariance between $r_3(2)$ and $r_7(2)$?
4. What is the conditional distribution of $r_t(2)$ given r_{t-1} ?

Question 2

Use the code below to import the data file **anz1.xlsx** into RStudio (make sure that you have saved the data into the same directory as your **.qmd** file in your R Project) and convert it to **xts** format. This file contains data on the price, volume, low price, high price, and closing price of the ANZ stock price from 2005 to 2007, at a daily frequency.

```
anz1.df <- readxl::read_excel("anz1.xlsx")

library(xts)
anz1.xts <- as.xts(anz1.df)
```

Now use the code below to calculate log returns (close-to-close) and then plot the ANZ share price and the log returns over the sample period.

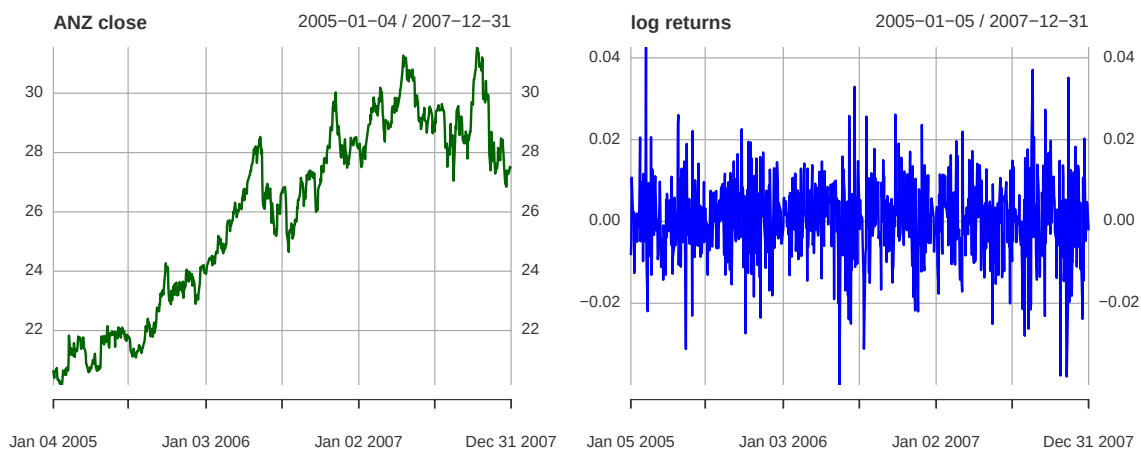
```
# Figure settings in r code header above:
# (fig.width=10, fig.height=4, dpi=150, fig.show='hold') stop the plot headings
# from overlapping. You may not need it all the time - only when printing
# multiple graphs

# log return for Close
logret <- diff(log(anz1.xts$CLOSE))
logret <- na.omit(logret) # removes the first NA created by diff()

# Plot 2 graphs in one row
# use mar = and mgp to adjust headings
par(mfrow = c(1, 2), mar = c(6, 4, 4, 2) + 0.1, mgp = c(2.2, 0.7, 0))

# xaxt = "n" also helps with headings
plot(anz1.xts$CLOSE, main = "ANZ close", col = "darkgreen", xaxt = "n")
plot(logret, main = "log returns", col = "blue")

# Reset layout
par(mfrow = c(1, 1))
```



We can see, as expected, that the close price series is not stationary, but the log-returns series is stationary.

One interesting point from the log-return series is that we can see some evidence of volatility clustering. This is where there are sustained periods of high variation (around early 2005, 2006 and near the end of the series in 2007).

1. Find the estimates of the first four sample moments of the returns. Do the returns exhibit evidence of stylised facts? Discuss.

Here we just need to use the descriptive statistics table for log returns from Week 2.

	Statistic	Log_Returns
1	Mean	0.0004
2	Median	0.0005
3	Maximum	0.0426
4	Minimum	-0.0399
5	Std_Dev	0.0102
6	Skewness	-0.1532
7	Excess Kurtosis	1.2928
8	Obs	756.0000

Answer here

2. Plot the histogram of returns on ANZ over this period. Add labels, colors and an appropriate number of breaks (or bins).

```
hist(logret, main = "??", breaks = ??, xlab = "??",
     col = "??", border = "black")
```

3. Comment on any deviations from normality suggested by these sample moments of returns.

Answer here

4. Conduct a formal hypothesis test, at the 5% significance level, that the mean of returns is zero, against the alternative that it is not zero.

We can simply run a regression of the log returns on a constant.

```
# we can use coredata so that we don't have to create a dataframe
# coredata() is from the zoo (and used by xts) package. It extracts the
# underlying data from a time-indexed object, stripping off the index
# (dates/times) and attributes, and returning a plain vector/matrix
# suitable for functions that expect standard R types.

fit <- lm(coredata(logret) ~ 1)
summary(fit)
```

```
Call:
lm(formula = coredata(logret) ~ 1)

Residuals:
    Min       1Q   Median       3Q      Max
-0.040309 -0.006152  0.000085  0.006238  0.042203

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.0003764   0.0003719   1.012   0.312

Residual standard error: 0.01023 on 755 degrees of freedom
```

Answer here

5. Using the Jarque-Bera test statistic, conduct a formal hypothesis test that returns are normal.

```
# Jarque Bera for log return
library(tseries)
jb_open <- jarque.bera.test(logret)

jb_df <- data.frame(
  Value = c(
    as.numeric(jb_open$statistic), # JB Stat
    jb_open$p.value                # p-value
  ),
  row.names = c("JB Stat", "p-value"),
  check.names = FALSE
)

# Optional formatting
jb_df$Value <- signif(jb_df$Value, 4)

jb_df
```

```
              Value
JB Stat 5.560e+01
p-value 8.431e-13
```

We test H_0 : returns are normal against H_1 : returns are not normal. We will use the JB test statistic, $JB = \frac{n}{6} \left((\hat{\mu}_3^s - 0)^2 + \frac{(\hat{\mu}_4^s - 3)^2}{4} \right)$ which has an asymptotic χ^2 distribution with

2 degrees of freedom, under H_0 . The calculated JB statistic is 55.6 and has a p-value of 0. At all the usual levels of significance (10%, 5% and 1%), we would reject the null in favour of the alternative. We would conclude that there is very strong evidence to say that daily returns for ANZ between 2005 and 2007 do not follow a Normal distribution.